C1.1 Enzymes and metabolism: data-based questions

NAME	DATE	TOTAL
		_____ / 8

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 A cofactor and an immobilised enzyme

8 MARKS

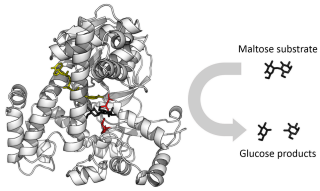
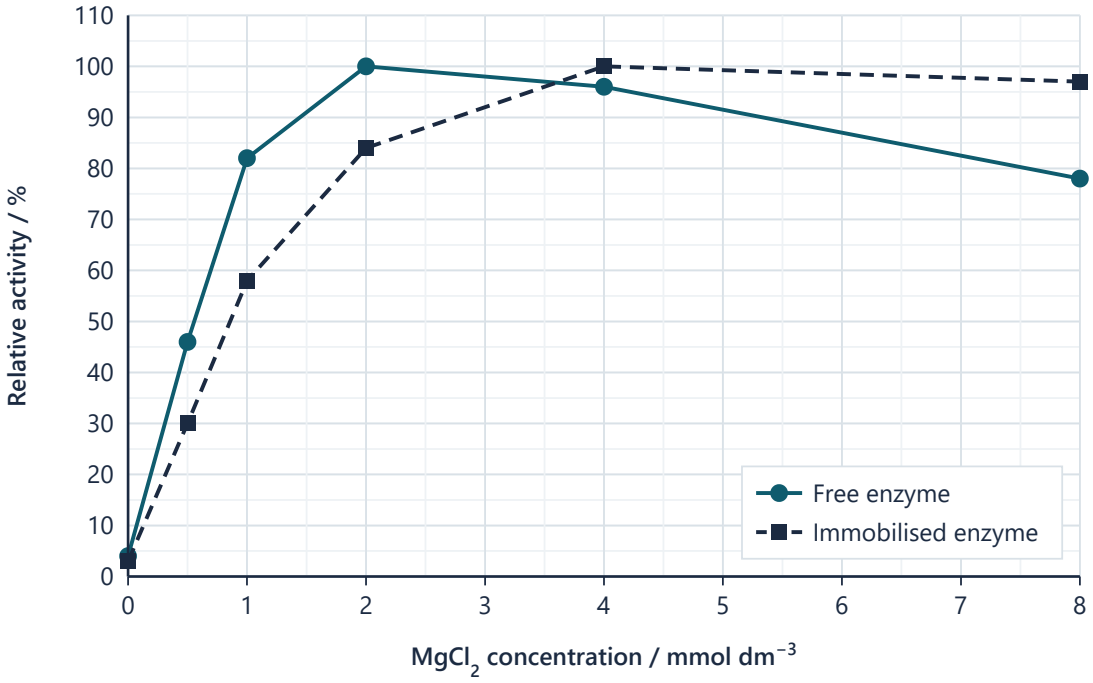


Image: Thomas Shafee, CC BY 4.0, Wikimedia Commons.

Hexokinase from yeast was tested free in solution and immobilised in alginate beads. ATP was kept at 1.0 mmol dm^{-3}, and the concentration of magnesium chloride ($MgCl_2$) was varied. Activity is given as a percentage of the highest activity measured for that form of the enzyme.



a STATE the relative activity of the free enzyme at 1.0 mmol dm^{-3} $MgCl_2$. [1]

b IDENTIFY the $MgCl_2$ concentration that gives the highest activity for each form of the enzyme. [2]

c CALCULATE the ratio of ATP to $MgCl_2$ concentration at which the free enzyme is most active. [1]

d EXPLAIN why there is almost no activity without magnesium ions. [2]

e SUGGEST why the immobilised enzyme needs a higher $MgCl_2$ concentration to reach its highest activity. [1]

f STATE one advantage of immobilising enzymes for use in industry. [1]